

Big Mart Sales Prediction Approach

The objective of this project is to develop a predictive model that estimates the sales of each product at a specific outlet.

The dataset contains two main categories of features:

Outlet-related attributes — such as location type, establishment year, outlet size, and outlet type.

Product-related attributes — such as maximum retail price (MRP), product weight, product type, and fat content.

These features play a crucial role in capturing the factors that influence how a product performs at different outlets. By leveraging both outlet characteristics and product details, the model aims to accurately forecast product-level sales across various store locations.

Methodology:

1. Exploratory Data Analysis (EDA)

A thorough univariate and bivariate analysis was conducted to understand the relationships between features and the target variable (Item_Outlet_Sales).

Key observations from the EDA include:

- Item MRP emerged as the most influential feature, showing a strong correlation with sales.
- The dataset contains a mix of categorical and numerical features. Among the categorical variables, both nominal and ordinal data types are present, making appropriate encoding strategies crucial.
- There were missing values in two key features: Item_Weight and Outlet_Size. Instead of dropping these records (due to the relatively small dataset and the importance of these variables), a KNN imputer was used to impute the missing values effectively.
- Some numerical features exhibited outliers and mild skewness, which were analyzed carefully to understand their potential impact on the model.
- An extensive analysis of how various categorical and numerical features influenced sales was also performed.

A detailed exploratory analysis can be found in the notebook: AAB_EDA.ipynb.

2. Feature Engineering:

Several feature engineering steps were performed to enhance model performance by providing additional context and improving data quality:

- **Missing Value Imputation:**
Missing values in the columns Item_Weight and Outlet_Size were imputed using KNN Imputer, which leverages patterns from similar data points to fill missing values more accurately.
- **Data Cleaning:**

The column `Item_Fat_Content` contained inconsistent labels (such as 'low fat', 'LF', 'Low Fat') which were standardized to ensure consistency.

- **Categorical Encoding:**

For ordinal features like `Outlet_Size` and `Outlet_Location_Type`, meaningful numerical values were assigned to preserve their inherent order. For instance, Tier 1 (a more urban location) was assigned a higher value compared to Tier 2 or Tier 3.

For nominal features (with no intrinsic order), one-hot encoding was applied to avoid introducing spurious ordinal relationships.

- **Item Visibility Correction:**

A significant portion of `Item_Visibility` values were recorded as zero, which is unrealistic. These zero values were imputed with the average visibility of the same product across different outlets. Additionally, a small constant factor was added to normalize for outlet size effects — ensuring visibility was not underestimated in larger outlets.

- **Derived Features:**

`Item_Type_Combined`: Extracted from the `Item_Identifier`, this feature grouped items into broader product categories (e.g., Food, Drinks, Non-Consumable), providing higher-level product insights.

`Outlet_Age`: Calculated as the age of the outlet as of 2013 (the latest year in the dataset). Older outlets may have stronger customer bases or different sales patterns.

`Sales_Frac`: Indicates the proportion of total products sold by a particular outlet, giving insight into outlet performance.

`Per_Unit_Cost`: Computed as the cost per unit weight of a product. This helped capture product affordability, which plays a role in consumer purchase behavior.

- `Essentials` and `Can_Spoil`: Binary indicators representing whether a product is a daily essential and whether it is perishable, respectively. These help model frequency of purchase and inventory turnover.
- `MRP_Category`: Categorizes products as cheap, moderate, or expensive relative to the average product price, providing further granularity into product pricing strategy.
- These engineered features provided richer information to the model and improved its ability to capture complex patterns in the data.

For more details please refer to ``ABB_Feature_Engineering.ipynb`` and ``ABB_Final_Model.ipynb``

3. Model training:

We experimented with various models — from simple regressions to neural networks and tree-based methods — to predict product sales.

Linear Regression Models:

Used as a baseline, but struggled with capturing complex, non-linear relationships, leading to limited performance.

Neural Networks:

Though capable of modeling non-linearities, they require large datasets and careful tuning. Given our limited data, they did not generalize well and underperformed.

Tree-Based Models (Final Model: RandomForestRegressor):

Given the mixed data types and complex feature interactions, tree-based models were a natural fit.

RandomForestRegressor outperformed other models due to:

- Ability to handle non-linearity, missing values, and mixed features
- Useful feature importance insights

After tuning hyperparameters using RandomizedSearchCV, this model delivered the best accuracy.

To further boost performance, we stacked a Polynomial Regression model on top of RandomForest predictions, reducing RMSE by ~2 points and improving leaderboard rank by ~50 positions.

For full implementation, see ABB_Final_Model.ipynb.